

Initiating Search

November 11, 2025, 2:18 PM

• Search:

CAS Lexicon Search:

Minisci reaction - Preferred Concept

Search Tasks

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Exported: Retrieved Related Reaction Results + Filters (262)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	

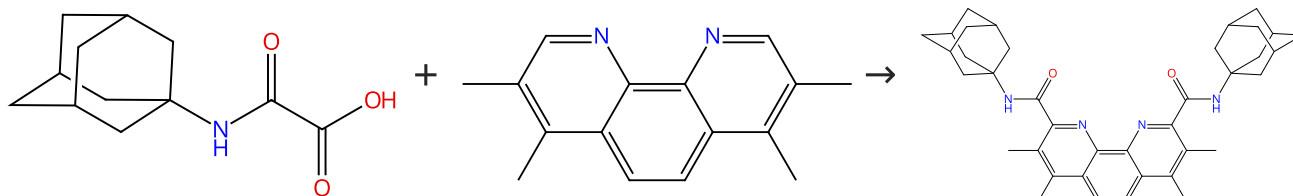
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 Reactions (50)[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%

 Suppliers (14) Suppliers (89)

31-614-CAS-41769675

Steps: 1 Yield: 100%

Direct C-H amidation of 1,3-azoles: light-mediated, photosensitizer-free vs. thermal

1.1 Reagents: Ammonium persulfate
Solvents: Dimethyl sulfoxide; 24 h, 25 °C

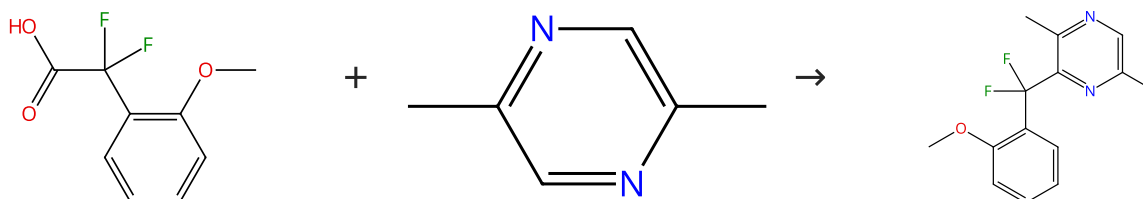
By: Mooney, David T.; et al

Experimental Protocols

Chemical Communications (Cambridge, United Kingdom) (2024), 60(77), 10752-10755.

Scheme 2 (1 Reaction)

Steps: 1 Yield: 99%

 Suppliers (47) Suppliers (92)

31-219-CAS-20990584

Steps: 1 Yield: 99%

Ag-Catalyzed Minisci C-H difluoromethylarylation of N-heteroarenes

1.1 Reagents: Ammonium persulfate
Catalysts: Silver nitrate
Solvents: Dichloromethane, Water; 24 h, 50 °C

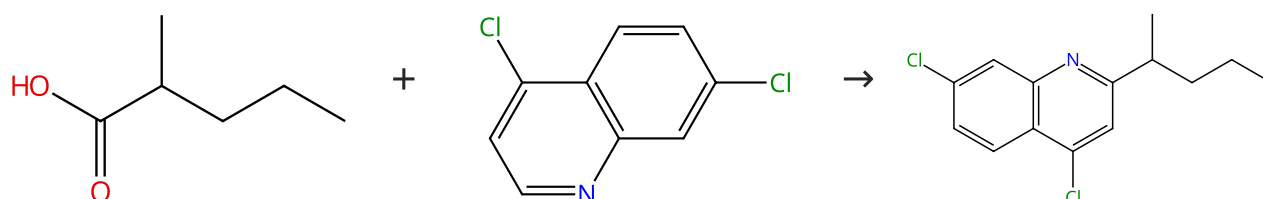
By: Xie, Xiaojuan; et al

Experimental Protocols

Organic & Biomolecular Chemistry (2020), 18(3), 400-404.

Scheme 3 (1 Reaction)

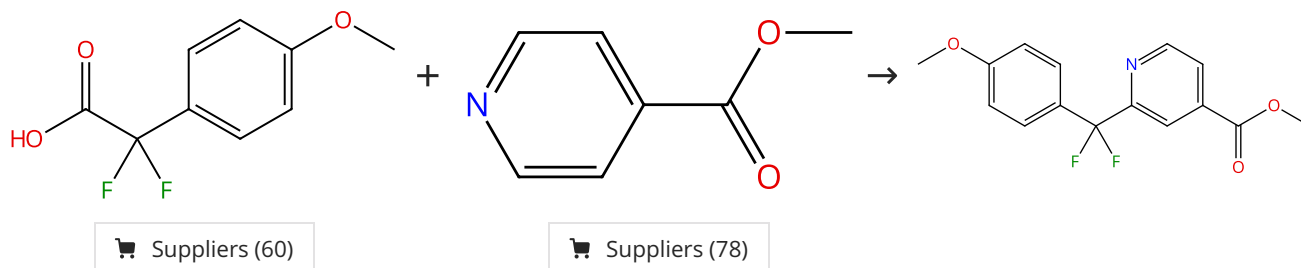
Steps: 1 Yield: 97%

 Suppliers (87) Suppliers (103)

31-219-CAS-19039939	Steps: 1 Yield: 97%	Photoredox-mediated Minisci alkylation of N-heteroarenes using carboxylic acids and hypervalent iodine
1.1 Reagents: 1-(Acetyloxy)-1,2-benziodoxol-3(1 <i>H</i>)-one Catalysts: Tris(2,2'-bipyridyl)ruthenium(II) chloride Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 8 h, 30 °C		By: Wang, Junhua; et al
1.2 Reagents: Potassium carbonate; 5 min		Asian Journal of Organic Chemistry (2018), 7(7), 1307-1310.
Experimental Protocols		

Scheme 4 (1 Reaction)

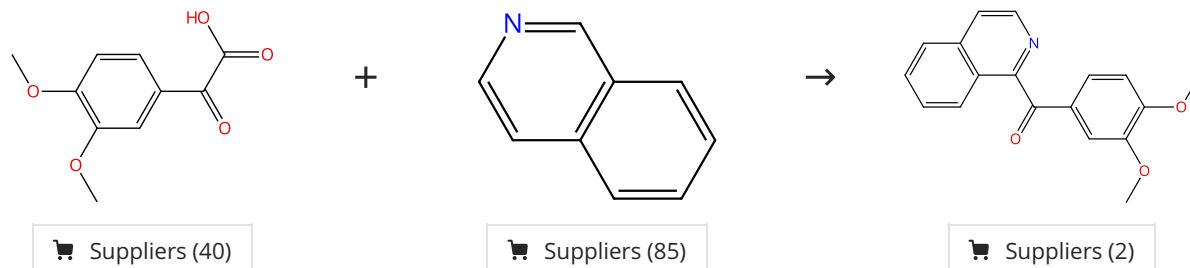
Steps: 1 Yield: 97%



31-219-CAS-20990603	Steps: 1 Yield: 97%	Ag-Catalyzed Minisci C-H difluoromethylarylation of N-heteroarenes
1.1 Reagents: Ammonium persulfate Catalysts: Silver nitrate Solvents: Dichloromethane, Water; 24 h, 50 °C		By: Xie, Xiaojuan; et al
Experimental Protocols		Organic & Biomolecular Chemistry (2020), 18(3), 400-404.

Scheme 5 (1 Reaction)

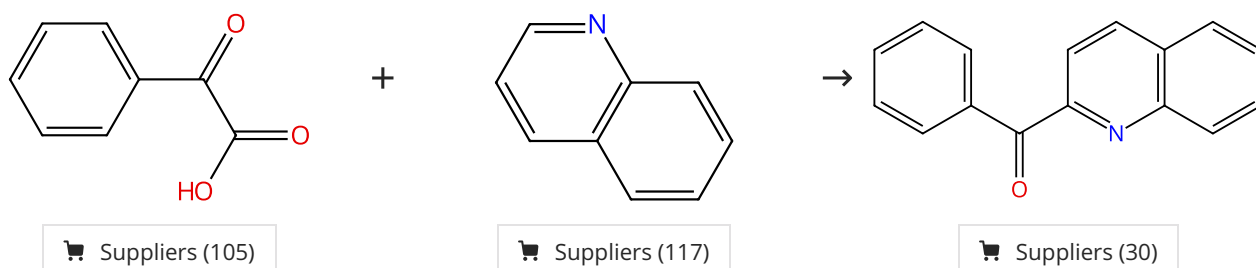
Steps: 1 Yield: 97%



31-614-CAS-24894214	Steps: 1 Yield: 97%	Minisci arylation of N-heterocycles using choline persulfate in water under mild conditions
1.1 Reagents: Potassium persulfate, Cholinium acetate Solvents: Water; 24 h, 40 °C		By: Laha, Joydev K.; et al
1.2 Reagents: Sodium bicarbonate Solvents: Water; neutralized		New Journal of Chemistry (2021), 45(48), 22853-22859.
Experimental Protocols		

Scheme 6 (1 Reaction)

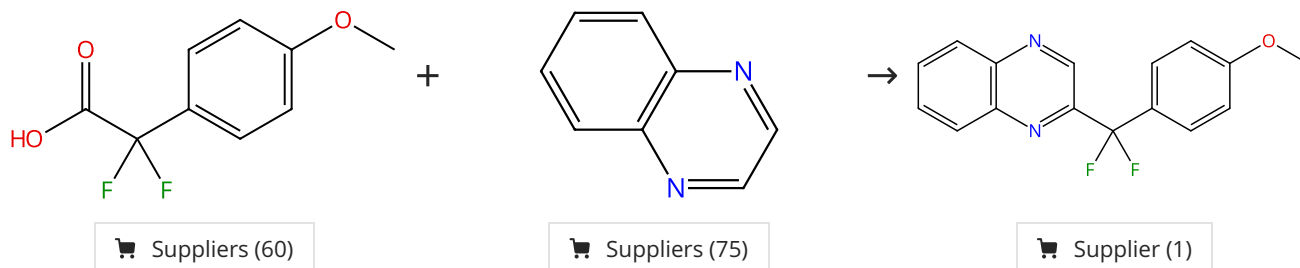
Steps: 1 Yield: 97%



31-614-CAS-24894186	Steps: 1 Yield: 97%	Minisci arylation of N-heterocycles using choline persulfate in water under mild conditions
1.1 Reagents: Potassium persulfate, Cholinium acetate Solvents: Water; 24 h, 40 °C		By: Laha, Joydev K.; et al
1.2 Reagents: Sodium bicarbonate Solvents: Water; neutralized		New Journal of Chemistry (2021), 45(48), 22853-22859.
Experimental Protocols		

Scheme 7 (1 Reaction)

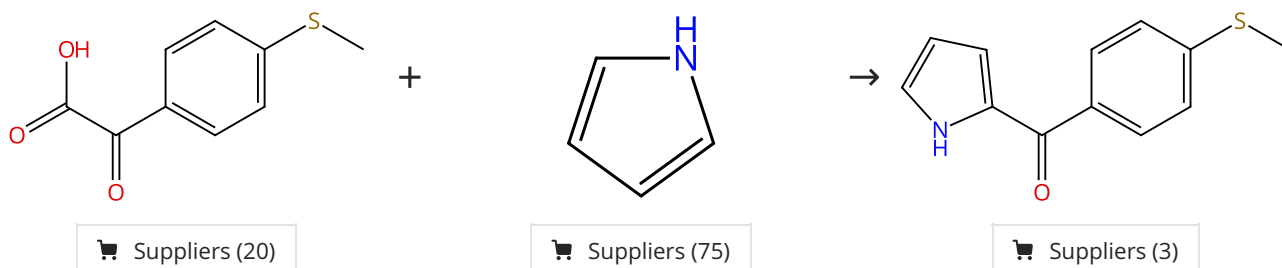
Steps: 1 Yield: 95%



31-219-CAS-20990598	Steps: 1 Yield: 95%	Ag-Catalyzed Minisci C-H difluoromethylarylation of N-hetero arenes
1.1 Reagents: Ammonium persulfate Catalysts: Silver nitrate Solvents: Dichloromethane, Water; 24 h, 50 °C		By: Xie, Xiaojuan; et al
Experimental Protocols		Organic & Biomolecular Chemistry (2020), 18(3), 400-404.

Scheme 8 (1 Reaction)

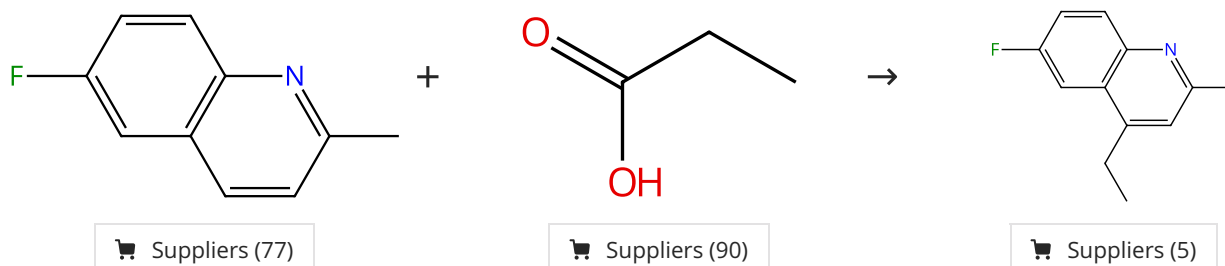
Steps: 1 Yield: 95%



31-219-CAS-21552246	Steps: 1 Yield: 95%	Arylation of Electron-Rich Pyrroles under Minisci Reaction Conditions
1.1 Reagents: Potassium persulfate Solvents: Acetonitrile; 6 h, 80 °C; 80 °C → rt		By: Laha, Joydev K.; et al
1.2 Reagents: Sodium carbonate Solvents: Water; neutralized		Organic Letters (2020), 22(4), 1442-1447.

Scheme 9 (1 Reaction)

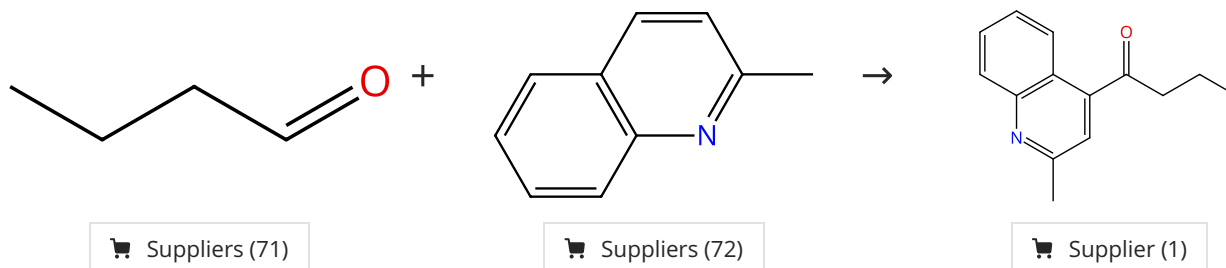
Steps: 1 Yield: 95%



31-614-CAS-36820342	Steps: 1 Yield: 95%	Ag-Catalyzed, Selectfluor-Mediated External Acid-Free Minisci Reaction in Aqueous Solution
1.1 Reagents: Selectfluor Catalysts: Silver nitrate Solvents: Water; 4 h, 80 °C		By: Wang, Wengui; et al
1.2 Reagents: Sodium bicarbonate Solvents: Water		ChemistrySelect (2023), 8(18), e202300958.
Experimental Protocols		

Scheme 10 (1 Reaction)

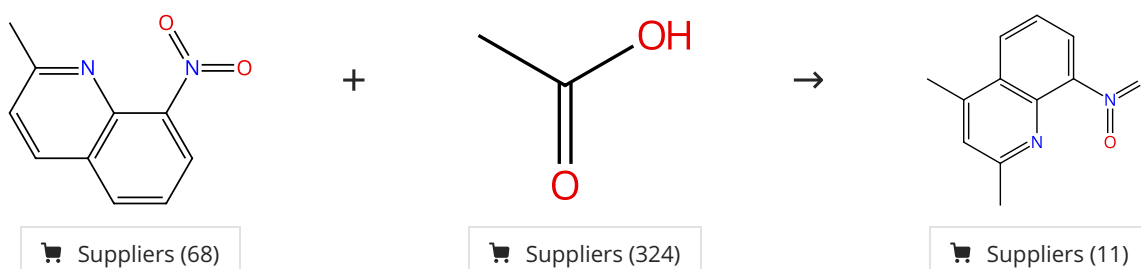
Steps: 1 Yield: 94%



31-614-CAS-36990071	Steps: 1 Yield: 94%	Ag-catalyzed acylation of N-heterocycles in aqueous solution
1.1 Reagents: Selectfluor Catalysts: Silver nitrate Solvents: Water; 24 h, 30 °C		By: Song, Yingyue; et al
Experimental Protocols		Tetrahedron (2023), 141, 133518.

Scheme 11 (1 Reaction)

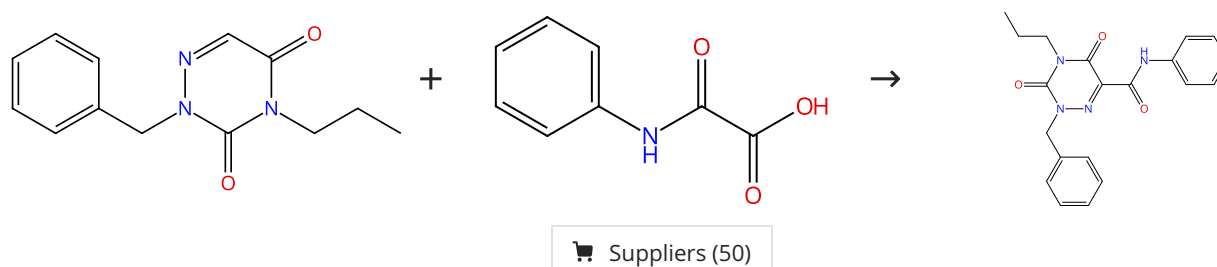
Steps: 1 Yield: 94%



31-614-CAS-36820340	Steps: 1 Yield: 94%	Ag-Catalyzed, Selectfluor-Mediated External Acid-Free Minisci Reaction in Aqueous Solution
1.1 Reagents: Selectfluor Catalysts: Silver nitrate Solvents: Water; 4 h, 80 °C		By: Wang, Wengui; et al
1.2 Reagents: Sodium bicarbonate Solvents: Water		ChemistrySelect (2023), 8(18), e202300958.
Experimental Protocols		

Scheme 12 (1 Reaction)

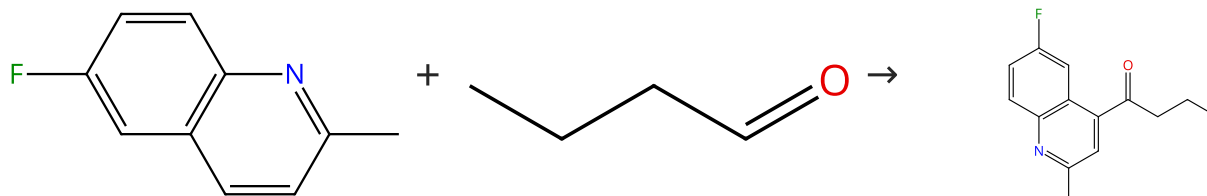
Steps: 1 Yield: 94%



31-614-CAS-39352069	Steps: 1 Yield: 94%	Minisci-Type Carbamoylation of Azauracils with Oxamic Acids
1.1 Reagents: Ammonium persulfate Solvents: Dimethyl sulfoxide; 12 h, 80 °C		By: Pan, Changduo; et al
Experimental Protocols		European Journal of Organic Chemistry (2024), 27(7), e202301177.

Scheme 13 (1 Reaction)

Steps: 1 Yield: 94%



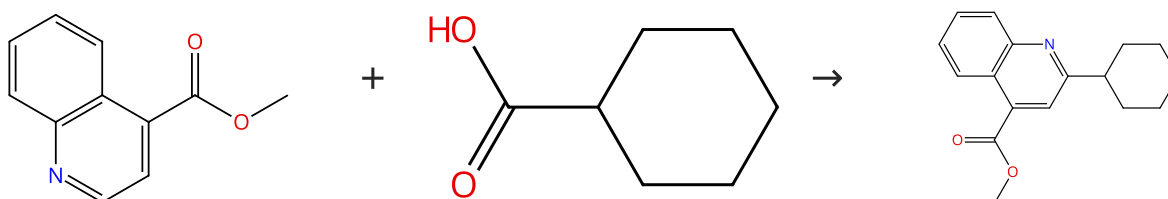
Suppliers (77)

Suppliers (71)

31-614-CAS-36990074	Steps: 1 Yield: 94%	Ag-catalyzed acylation of N-heterocycles in aqueous solution
1.1 Reagents: Selectfluor Catalysts: Silver nitrate Solvents: Water; 24 h, 30 °C		By: Song, Yingyue; et al
Experimental Protocols		Tetrahedron (2023), 141, 133518.

Scheme 14 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (62)

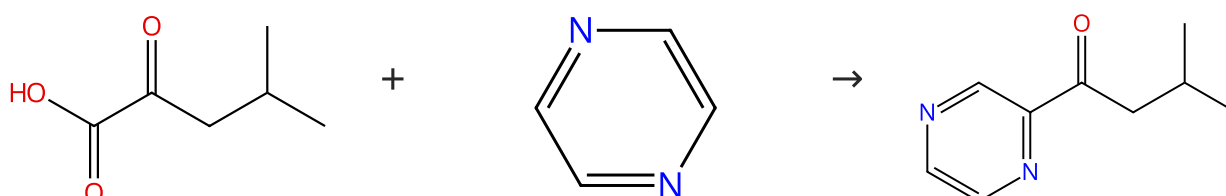
Suppliers (96)

Suppliers (2)

31-219-CAS-19039950	Steps: 1 Yield: 93%	Photoredox-mediated Minisci alkylation of N-heteroarenes using carboxylic acids and hypervalent iodine
1.1 Reagents: 1-(Acetyloxy)-1,2-benziodoxol-3(1 <i>H</i>)-one Catalysts: Tris(2,2'-bipyridyl)ruthenium(II) chloride Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 8 h, 30 °C		By: Wang, Junhua; et al
1.2 Reagents: Potassium carbonate; 5 min		Asian Journal of Organic Chemistry (2018), 7(7), 1307-1310.
Experimental Protocols		

Scheme 15 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (71)

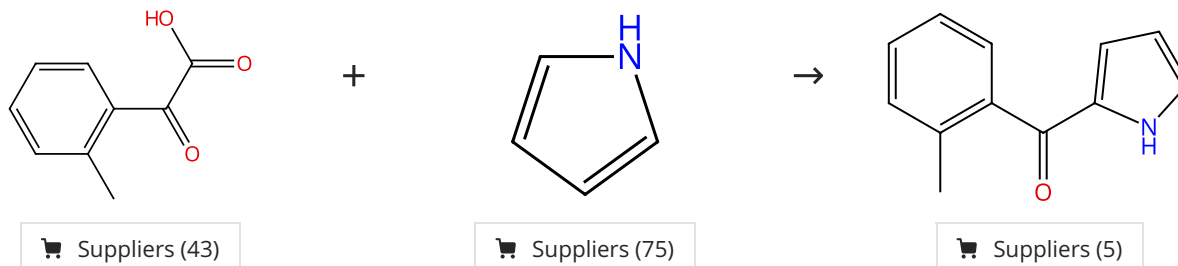
Suppliers (91)

Suppliers (8)

31-219-CAS-19884998 Steps: 1 Yield: 93% 1.1 Reagents: Formic acid, Ammonium thiosulfate Catalysts: Ferrous sulfate heptahydrate Solvents: Dimethyl sulfoxide, Dichloromethane, Water; 40 °C 1.2 Reagents: Sodium hydroxide Solvents: Water; neutralized Experimental Protocols	Iron-catalyzed Minisci acylation of N-heteroarenes with α-keto acids By: Wang, Xiu-Zhi; et al Tetrahedron (2019), 75(10), 1425-1430.
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Scheme 16 (1 Reaction)

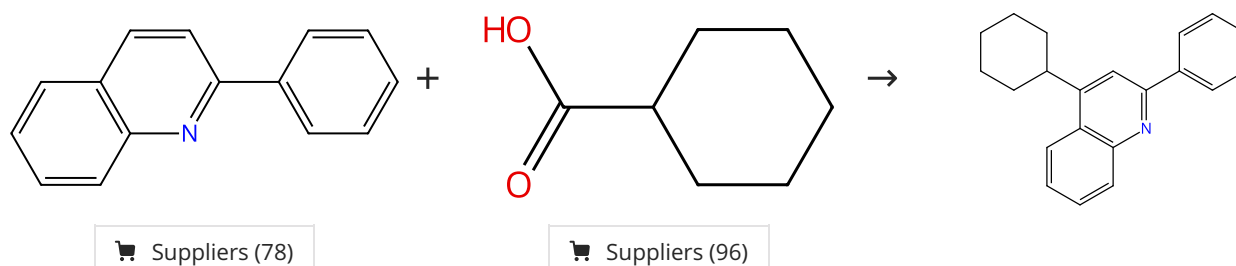
Steps: 1 Yield: 93%



31-219-CAS-21552237 Steps: 1 Yield: 93% 1.1 Reagents: Potassium persulfate Solvents: Acetonitrile; 6 h, 80 °C; 80 °C → rt 1.2 Reagents: Sodium carbonate Solvents: Water; neutralized	Aroylation of Electron-Rich Pyrroles under Minisci Reaction Conditions By: Laha, Joydev K.; et al Organic Letters (2020), 22(4), 1442-1447.
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Scheme 17 (1 Reaction)

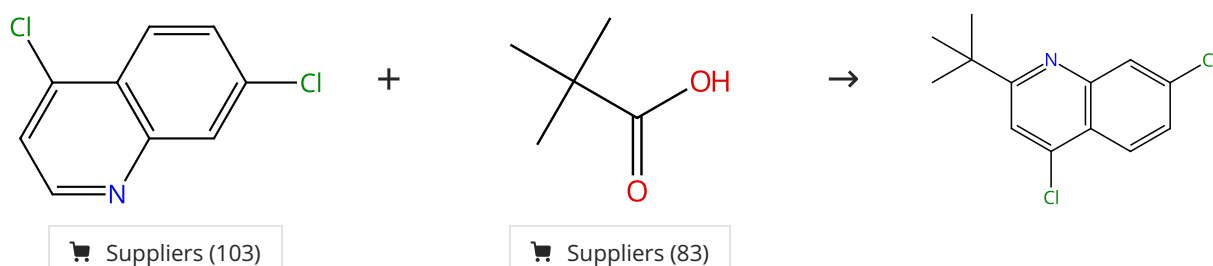
Steps: 1 Yield: 93%



31-219-CAS-19039947 Steps: 1 Yield: 93% 1.1 Reagents: 1-(Acetyloxy)-1,2-benziodoxol-3(1<i>H</i>)-one Catalysts: Tris(2,2'-bipyridyl)ruthenium(II) chloride Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 15 h, 30 °C 1.2 Reagents: Potassium carbonate; 5 min Experimental Protocols	Photoredox-mediated Minisci alkylation of N-heteroarenes using carboxylic acids and hypervalent iodine By: Wang, Junhua; et al Asian Journal of Organic Chemistry (2018), 7(7), 1307-1310.
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Scheme 18 (1 Reaction)

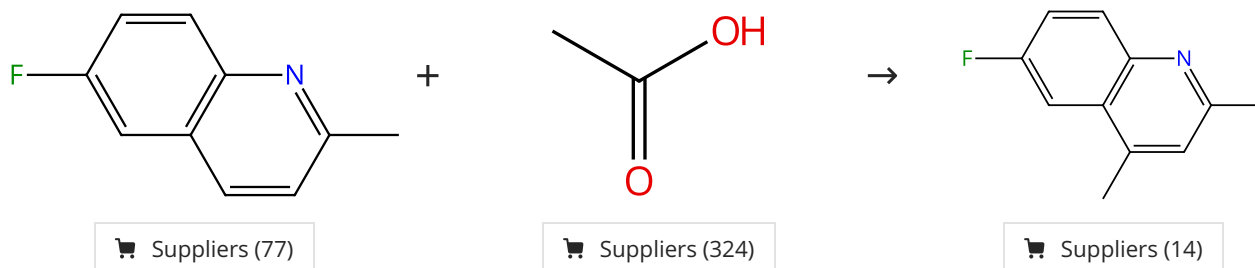
Steps: 1 Yield: 93%



31-219-CAS-19039940	Steps: 1 Yield: 93%	Photoredox-mediated Minisci alkylation of N-heteroarenes using carboxylic acids and hypervalent iodine
1.1 Reagents: 1-(Acetyloxy)-1,2-benziodoxol-3(1 <i>H</i>)-one Catalysts: Tris(2,2'-bipyridyl)ruthenium(II) chloride Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 8 h, 30 °C		By: Wang, Junhua; et al
1.2 Reagents: Potassium carbonate; 5 min		Asian Journal of Organic Chemistry (2018), 7(7), 1307-1310.
Experimental Protocols		

Scheme 19 (1 Reaction)

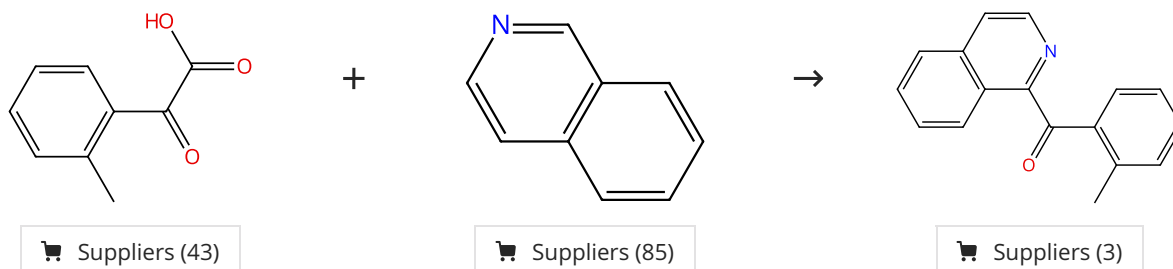
Steps: 1 Yield: 93%



31-614-CAS-36820343	Steps: 1 Yield: 93%	Ag-Catalyzed, Selectfluor-Mediated External Acid-Free Minisci Reaction in Aqueous Solution
1.1 Reagents: Selectfluor Catalysts: Silver nitrate Solvents: Water; 4 h, 80 °C		By: Wang, Wengui; et al
1.2 Reagents: Sodium bicarbonate Solvents: Water		ChemistrySelect (2023), 8(18), e202300958.
Experimental Protocols		

Scheme 20 (1 Reaction)

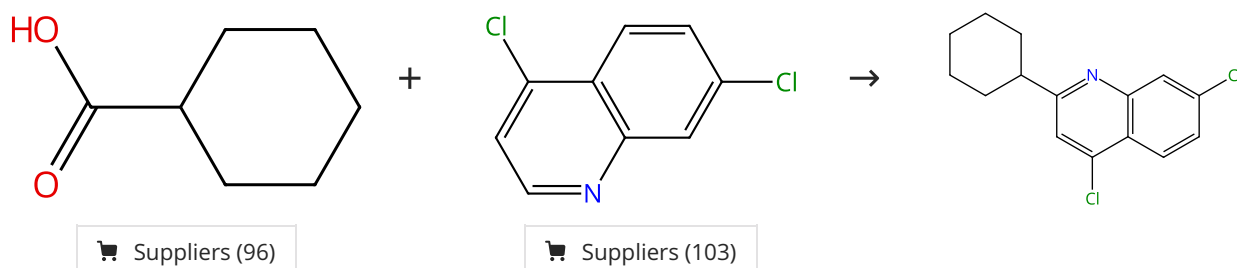
Steps: 1 Yield: 93%



31-614-CAS-24894207	Steps: 1 Yield: 93%	Minisci arylation of N-heterocycles using choline persulfate in water under mild conditions
1.1 Reagents: Potassium persulfate, Cholinium acetate Solvents: Water; 24 h, 40 °C		By: Laha, Joydev K.; et al
1.2 Reagents: Sodium bicarbonate Solvents: Water; neutralized		New Journal of Chemistry (2021), 45(48), 22853-22859.
Experimental Protocols		

Scheme 21 (1 Reaction)

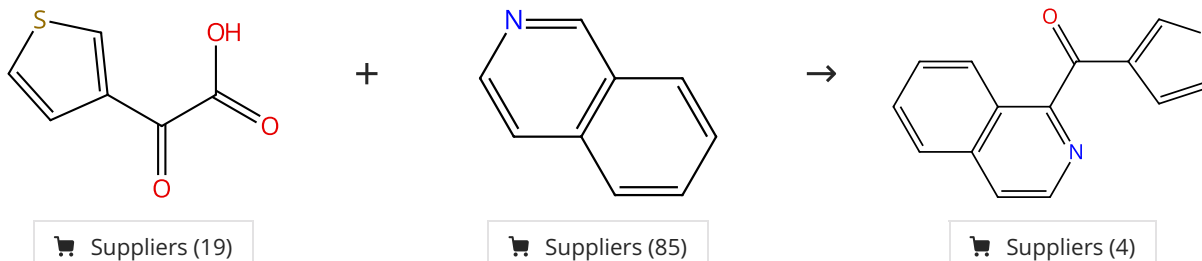
Steps: 1 Yield: 93%



31-219-CAS-19039931 Steps: 1 Yield: 93% 1.1 Reagents: 1-(Acetyloxy)-1,2-benziodoxol-3(1<i>H</i>)-one Catalysts: Tris(2,2'-bipyridyl)ruthenium(II) chloride Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 8 h, 30 °C 1.2 Reagents: Potassium carbonate; 5 min Experimental Protocols	Photoredox-mediated Minisci alkylation of N-heteroarenes using carboxylic acids and hypervalent iodine By: Wang, Junhua; et al Asian Journal of Organic Chemistry (2018), 7(7), 1307-1310.
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Scheme 22 (1 Reaction)

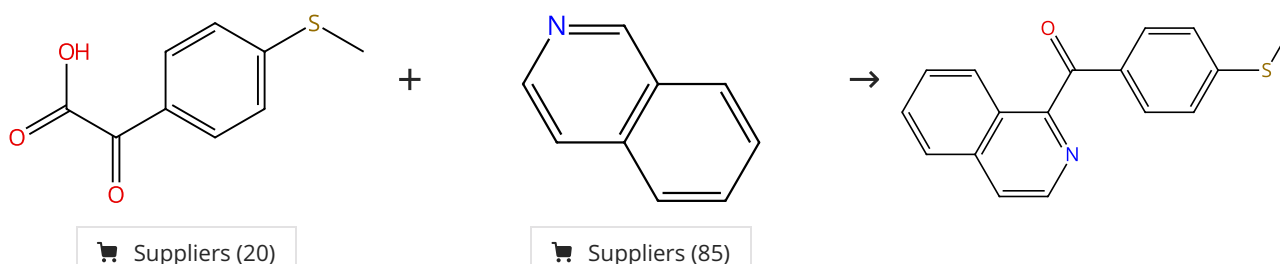
Steps: 1 Yield: 92%



31-614-CAS-24894215 Steps: 1 Yield: 92% 1.1 Reagents: Potassium persulfate, Cholinium acetate Solvents: Water; 24 h, 40 °C 1.2 Reagents: Sodium bicarbonate Solvents: Water; neutralized Experimental Protocols	Minisci arylation of N-heterocycles using choline persulfate in water under mild conditions By: Laha, Joydev K.; et al New Journal of Chemistry (2021), 45(48), 22853-22859.
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Scheme 23 (1 Reaction)

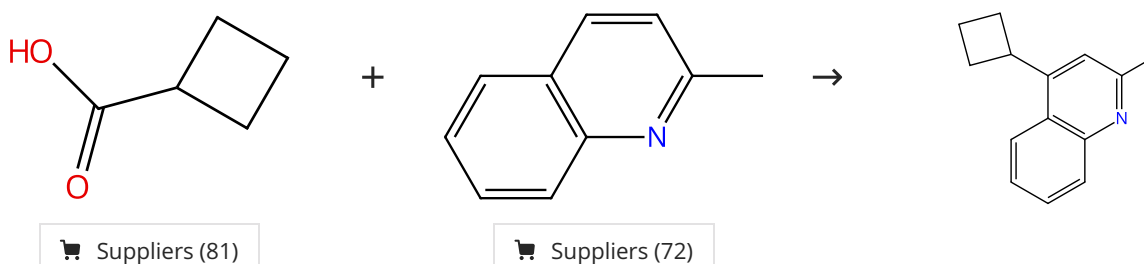
Steps: 1 Yield: 92%



31-614-CAS-24894197 Steps: 1 Yield: 92% 1.1 Reagents: Potassium persulfate, Cholinium acetate Solvents: Water; 24 h, 40 °C 1.2 Reagents: Sodium bicarbonate Solvents: Water; neutralized Experimental Protocols	Minisci arylation of N-heterocycles using choline persulfate in water under mild conditions By: Laha, Joydev K.; et al New Journal of Chemistry (2021), 45(48), 22853-22859.
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Scheme 24 (1 Reaction)

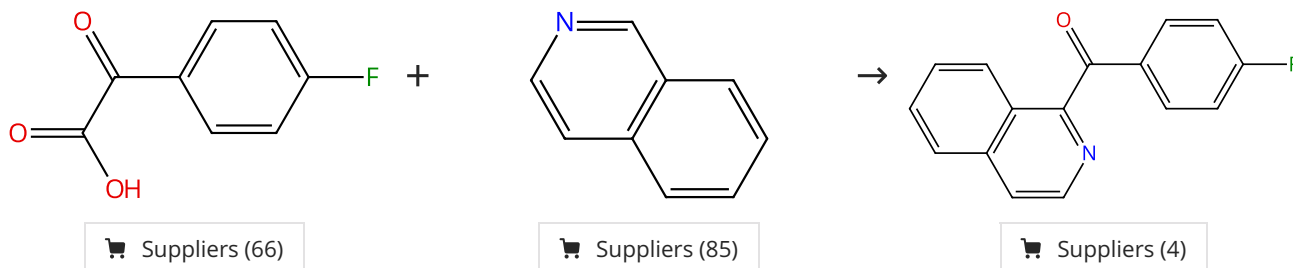
Steps: 1 Yield: 92%



31-614-CAS-36820366	Steps: 1 Yield: 92%	Ag-Catalyzed, Selectfluor-Mediated External Acid-Free Minisci Reaction in Aqueous Solution
1.1 Reagents: Selectfluor Catalysts: Silver nitrate Solvents: Water; 4 h, 80 °C		By: Wang, Wengui; et al
1.2 Reagents: Sodium bicarbonate Solvents: Water		ChemistrySelect (2023), 8(18), e202300958.
Experimental Protocols		

Scheme 25 (1 Reaction)

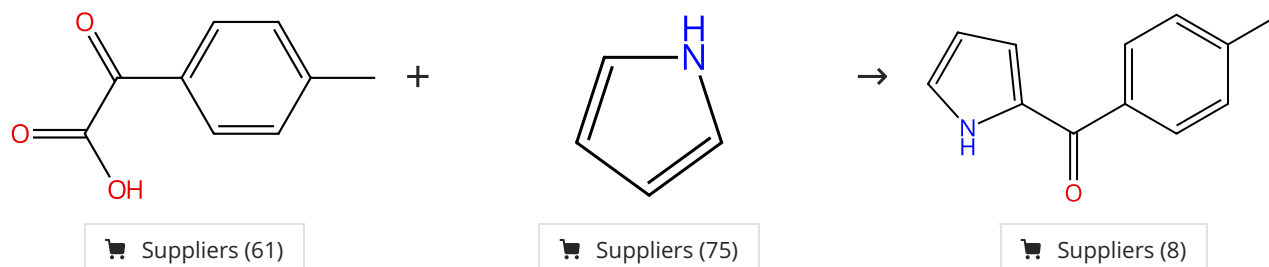
Steps: 1 Yield: 92%



31-614-CAS-24894205	Steps: 1 Yield: 92%	Minisci arylation of N-heterocycles using choline persulfate in water under mild conditions
1.1 Reagents: Potassium persulfate, Cholinium acetate Solvents: Water; 24 h, 40 °C		By: Laha, Joydev K.; et al
1.2 Reagents: Sodium bicarbonate Solvents: Water; neutralized		New Journal of Chemistry (2021), 45(48), 22853-22859.
Experimental Protocols		

Scheme 26 (1 Reaction)

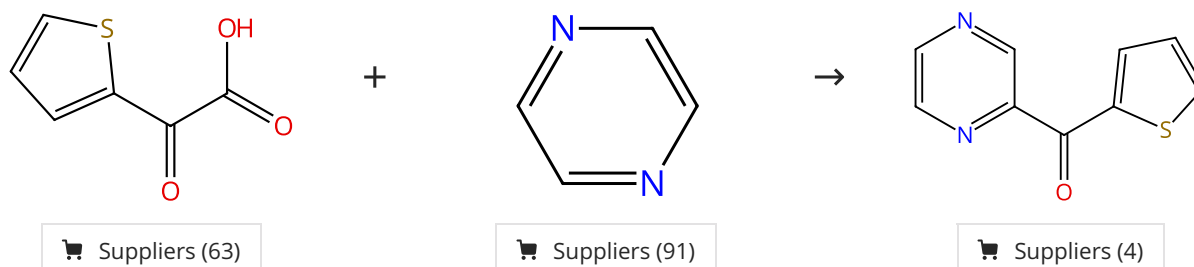
Steps: 1 Yield: 92%



31-219-CAS-21552247	Steps: 1 Yield: 92%	Arylation of Electron-Rich Pyrroles under Minisci Reaction Conditions
1.1 Reagents: Potassium persulfate Solvents: Acetonitrile; 6 h, 80 °C; 80 °C → rt		By: Laha, Joydev K.; et al
1.2 Reagents: Sodium carbonate Solvents: Water; neutralized		Organic Letters (2020), 22(4), 1442-1447.

Scheme 27 (1 Reaction)

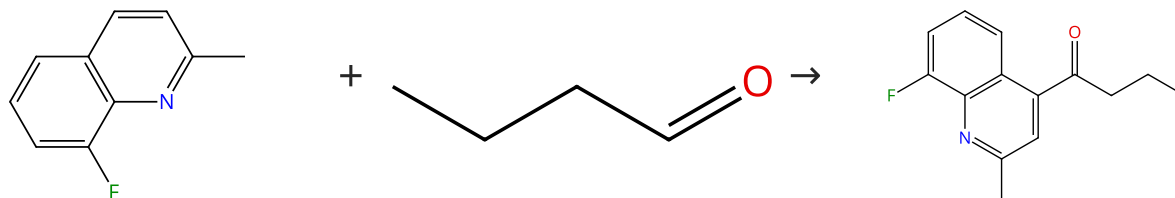
Steps: 1 Yield: 92%



31-219-CAS-19885012	Steps: 1 Yield: 92%	Iron-catalyzed Minisci acylation of N-heteroarenes with α-keto acids
1.1 Reagents: Formic acid, Ammonium thiosulfate Catalysts: Ferrous sulfate heptahydrate Solvents: Dimethyl sulfoxide, Dichloromethane, Water; 40 °C		By: Wang, Xiu-Zhi; et al
1.2 Reagents: Sodium hydroxide Solvents: Water; neutralized		Tetrahedron (2019), 75(10), 1425-1430.
Experimental Protocols		

Scheme 28 (1 Reaction)

Steps: 1 Yield: 92%



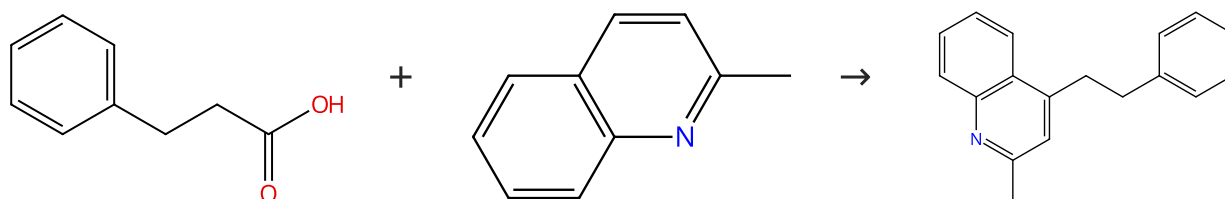
Suppliers (48)

Suppliers (71)

31-614-CAS-36990081	Steps: 1 Yield: 92%	Ag-catalyzed acylation of N-heterocycles in aqueous solution
1.1 Reagents: Selectfluor Catalysts: Silver nitrate Solvents: Water; 24 h, 30 °C		By: Song, Yingyue; et al
		Tetrahedron (2023), 141, 133518.
Experimental Protocols		

Scheme 29 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (107)

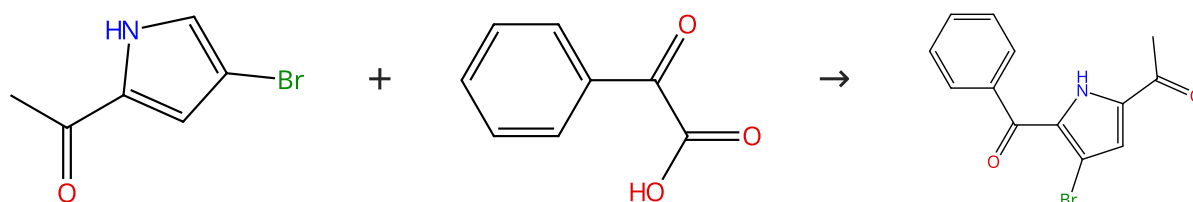
Suppliers (72)

Suppliers (3)

31-614-CAS-36820353	Steps: 1 Yield: 92%	Ag-Catalyzed, Selectfluor-Mediated External Acid-Free Minisci Reaction in Aqueous Solution
1.1 Reagents: Selectfluor Catalysts: Silver nitrate Solvents: Water; 4 h, 80 °C		By: Wang, Wengui; et al
1.2 Reagents: Sodium bicarbonate Solvents: Water		ChemistrySelect (2023), 8(18), e202300958.
Experimental Protocols		

Scheme 30 (1 Reaction)

Steps: 1 Yield: 92%



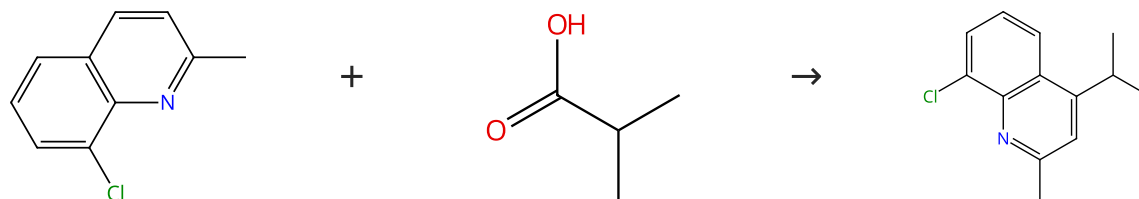
Suppliers (44)

Suppliers (105)

31-219-CAS-21552253	Steps: 1 Yield: 92%	Aroylation of Electron-Rich Pyrroles under Minisci Reaction Conditions
1.1 Reagents: Potassium persulfate Solvents: Acetonitrile; 6 h, 80 °C; 80 °C → rt		By: Laha, Joydev K.; et al
1.2 Reagents: Sodium carbonate Solvents: Water; neutralized		Organic Letters (2020), 22(4), 1442-1447.

Scheme 31 (1 Reaction)

Steps: 1 Yield: 92%



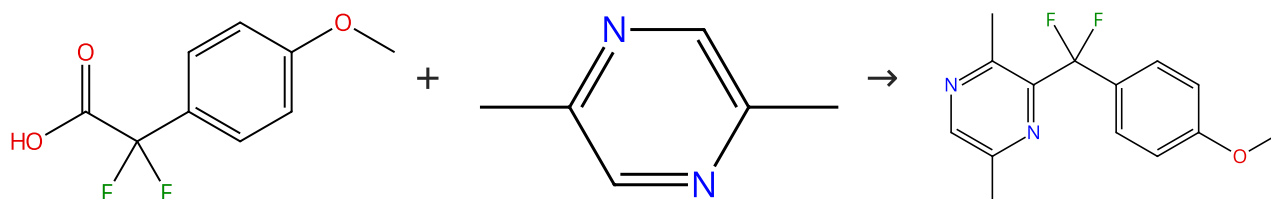
Suppliers (65)

Suppliers (77)

31-614-CAS-36820359	Steps: 1 Yield: 92%	Ag-Catalyzed, Selectfluor-Mediated External Acid-Free Minisci Reaction in Aqueous Solution
1.1 Reagents: Selectfluor Catalysts: Silver nitrate Solvents: Water; 4 h, 80 °C		By: Wang, Wengui; et al
1.2 Reagents: Sodium bicarbonate Solvents: Water		ChemistrySelect (2023), 8(18), e202300958.
Experimental Protocols		

Scheme 32 (1 Reaction)

Steps: 1 Yield: 92%



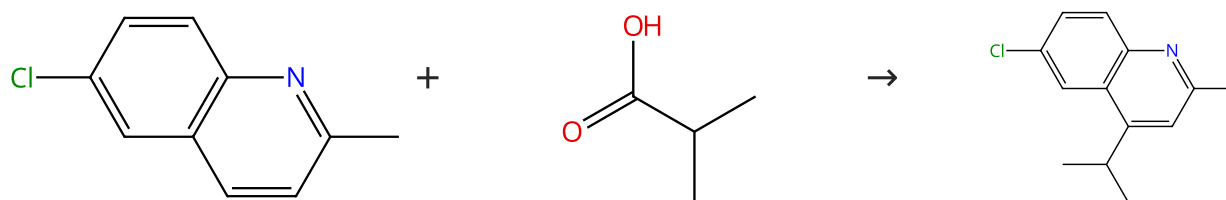
Suppliers (60)

Suppliers (92)

31-219-CAS-20990583	Steps: 1 Yield: 92%	Ag-Catalyzed Minisci C-H difluoromethylarylation of N-hetero arenes
1.1 Reagents: Ammonium persulfate Catalysts: Silver nitrate Solvents: Dichloromethane, Water; 24 h, 50 °C		By: Xie, Xiaojuan; et al
Experimental Protocols		Organic & Biomolecular Chemistry (2020), 18(3), 400-404.

Scheme 33 (1 Reaction)

Steps: 1 Yield: 92%



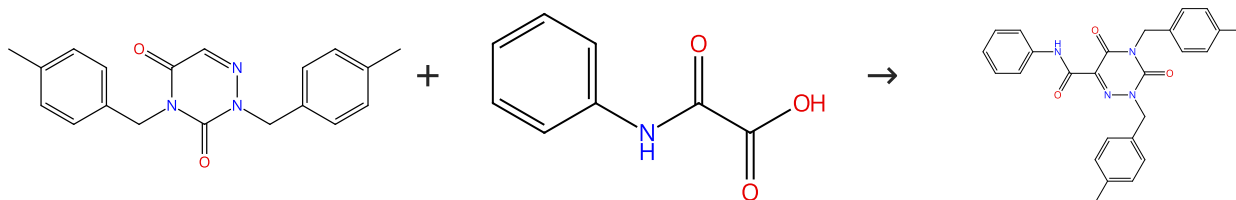
Suppliers (69)

Suppliers (77)

31-614-CAS-36820356	Steps: 1 Yield: 92%	Ag-Catalyzed, Selectfluor-Mediated External Acid-Free Minisci Reaction in Aqueous Solution
1.1 Reagents: Selectfluor Catalysts: Silver nitrate Solvents: Water; 4 h, 80 °C		By: Wang, Wengui; et al
1.2 Reagents: Sodium bicarbonate Solvents: Water		ChemistrySelect (2023), 8(18), e202300958.
Experimental Protocols		

Scheme 34 (1 Reaction)

Steps: 1 Yield: 91%

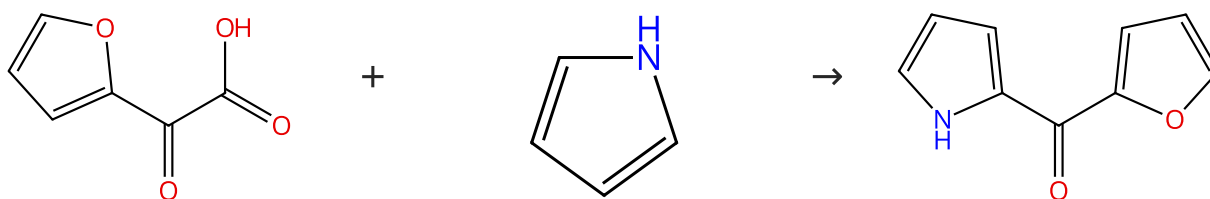


Suppliers (50)

31-614-CAS-39352065	Steps: 1 Yield: 91%	Minisci-Type Carbamoylation of Azauracils with Oxamic Acids
1.1 Reagents: Ammonium persulfate Solvents: Dimethyl sulfoxide; 12 h, 80 °C		By: Pan, Changduo; et al
Experimental Protocols		European Journal of Organic Chemistry (2024), 27(7), e202301177.

Scheme 35 (1 Reaction)

Steps: 1 Yield: 91%



Suppliers (60)

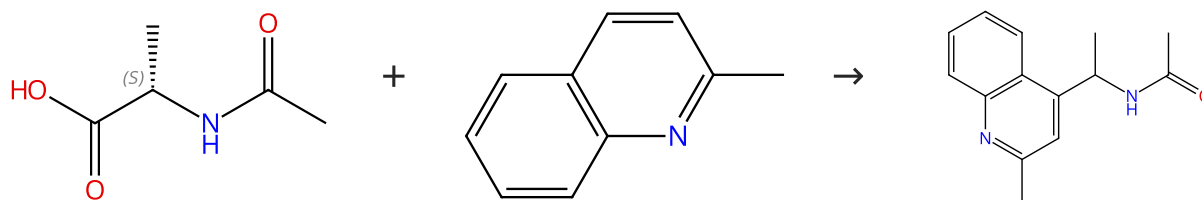
Suppliers (75)

Suppliers (6)

31-219-CAS-21552250	Steps: 1 Yield: 91%	Aroylation of Electron-Rich Pyrroles under Minisci Reaction Conditions
1.1 Reagents: Potassium persulfate Solvents: Acetonitrile; 6 h, 80 °C; 80 °C → rt		By: Laha, Joydev K.; et al
1.2 Reagents: Sodium carbonate Solvents: Water; neutralized		Organic Letters (2020), 22(4), 1442-1447.

Scheme 36 (1 Reaction)

Steps: 1 Yield: 91%


 Absolute stereochemistry shown,
Rotation (-)

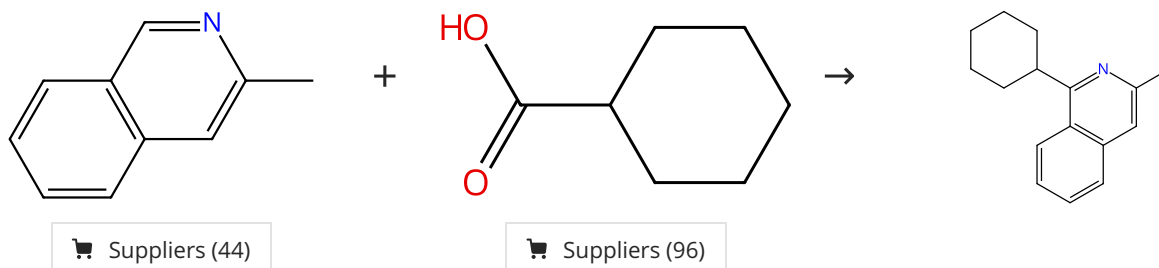
Suppliers (72)

Suppliers (87)

31-219-CAS-19352994	Steps: 1 Yield: 91%	Metal-, Photocatalyst-, and Light-Free, Late-Stage C-H Alkylation of Heteroarenes and 1,4-Quinones Using Carboxylic Acids By: Sutherland, Daniel R.; et al Organic Letters (2018), 20(21), 6863-6867.
1.1 Reagents: Ammonium persulfate Solvents: Dimethyl sulfoxide, Water; 5 min; overnight, 40 °C		
Experimental Protocols		

Scheme 37 (1 Reaction)

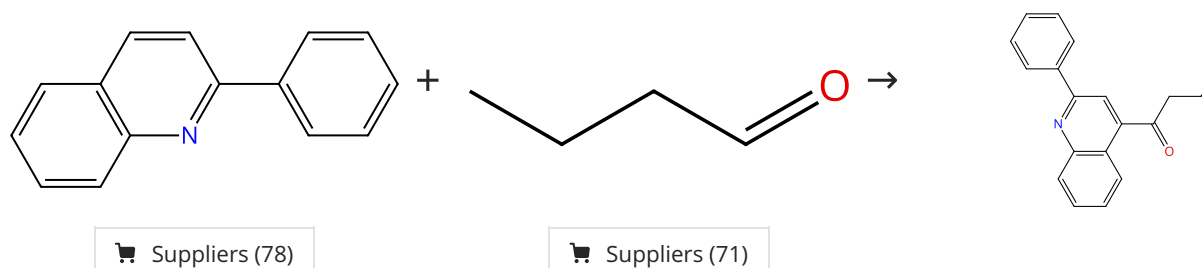
Steps: 1 Yield: 91%



31-219-CAS-19352983	Steps: 1 Yield: 91%	Metal-, Photocatalyst-, and Light-Free, Late-Stage C-H Alkylation of Heteroarenes and 1,4-Quinones Using Carboxylic Acids By: Sutherland, Daniel R.; et al Organic Letters (2018), 20(21), 6863-6867.
1.1 Reagents: Ammonium persulfate Solvents: Dimethyl sulfoxide, Water; 5 min; overnight, 40 °C		
Experimental Protocols		

Scheme 38 (1 Reaction)

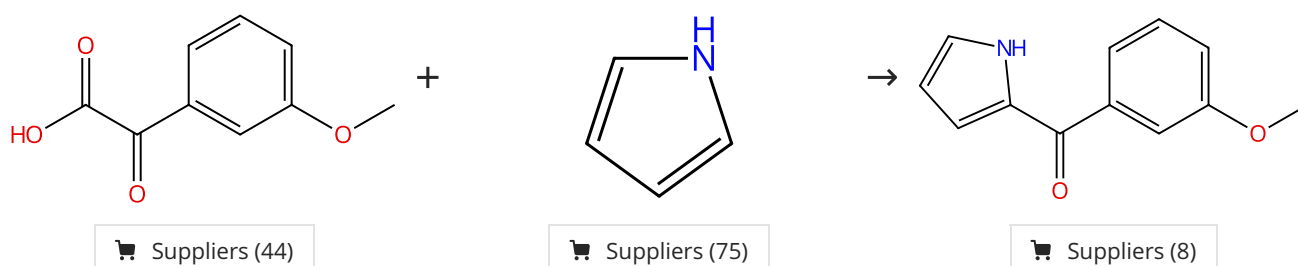
Steps: 1 Yield: 91%



31-614-CAS-39572394	Steps: 1 Yield: 91%	Minisci-type dehydrogenative coupling of N-heteroaromatic rings with inert C(sp³)-H enabled by visible light-catalyzed intermolecular hydrogen atom transfer process By: Yang, Jun-Feng; et al Journal of Organic Chemistry (2024), 89(6), 4249-4260.
1.1 Reagents: Trifluoroacetic acid, Benzoyl peroxide Solvents: Acetonitrile; 24 h, rt		
Experimental Protocols		

Scheme 39 (1 Reaction)

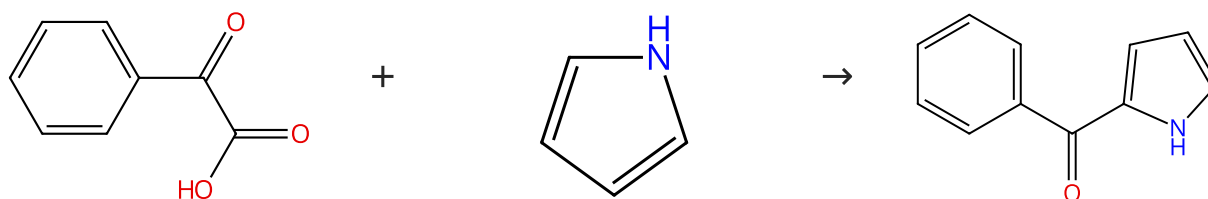
Steps: 1 Yield: 91%



31-219-CAS-21552242 Steps: 1 Yield: 91% 1.1 Reagents: Potassium persulfate Solvents: Acetonitrile; 6 h, 80 °C; 80 °C → rt 1.2 Reagents: Sodium carbonate Solvents: Water; neutralized	Aroylation of Electron-Rich Pyrroles under Minisci Reaction Conditions By: Laha, Joydev K.; et al Organic Letters (2020), 22(4), 1442-1447.
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Scheme 40 (1 Reaction)

Steps: 1 Yield: 91%



Suppliers (105)

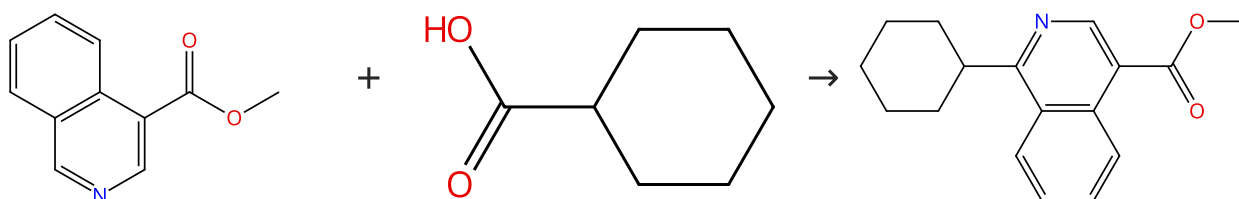
Suppliers (75)

Suppliers (67)

31-219-CAS-21552236 Steps: 1 Yield: 91% 1.1 Reagents: Potassium persulfate Solvents: Acetonitrile; 6 h, 80 °C; 80 °C → rt 1.2 Reagents: Sodium carbonate Solvents: Water; neutralized	Aroylation of Electron-Rich Pyrroles under Minisci Reaction Conditions By: Laha, Joydev K.; et al Organic Letters (2020), 22(4), 1442-1447.
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Scheme 41 (1 Reaction)

Steps: 1 Yield: 91%



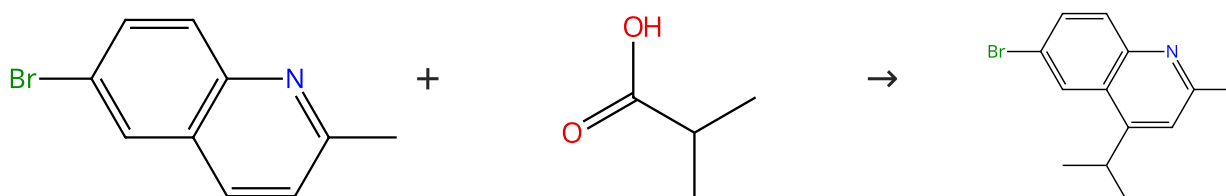
Suppliers (58)

Suppliers (96)

31-219-CAS-19039952 Steps: 1 Yield: 91% 1.1 Reagents: 1-(Acetyloxy)-1,2-benziodoxol-3(1H)-one Catalysts: Tris(2,2'-bipyridyl)ruthenium(II) chloride Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 8 h, 30 °C 1.2 Reagents: Potassium carbonate; 5 min Experimental Protocols	Photoredox-mediated Minisci alkylation of N-heteroarenes using carboxylic acids and hypervalent iodine By: Wang, Junhua; et al Asian Journal of Organic Chemistry (2018), 7(7), 1307-1310.
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Scheme 42 (1 Reaction)

Steps: 1 Yield: 91%



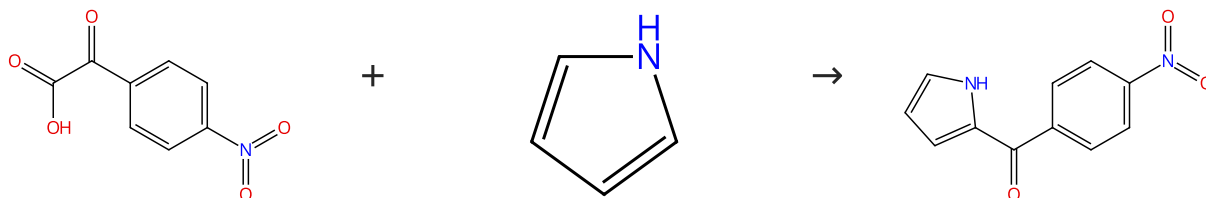
Suppliers (75)

Suppliers (77)

31-614-CAS-36820368	Steps: 1 Yield: 91%	Ag-Catalyzed, Selectfluor-Mediated External Acid-Free Minisci Reaction in Aqueous Solution
1.1 Reagents: Selectfluor Catalysts: Silver nitrate Solvents: Water; 4 h, 80 °C		By: Wang, Wengui; et al
1.2 Reagents: Sodium bicarbonate Solvents: Water		ChemistrySelect (2023), 8(18), e202300958.
Experimental Protocols		

Scheme 43 (1 Reaction)

Steps: 1 Yield: 90%



Suppliers (59)

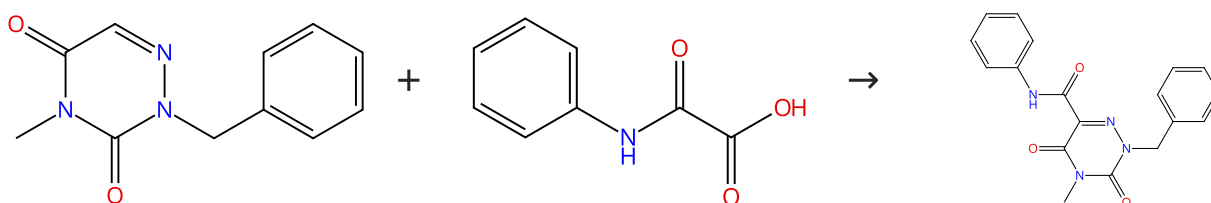
Suppliers (75)

Suppliers (19)

31-219-CAS-21552245	Steps: 1 Yield: 90%	Aroylation of Electron-Rich Pyrroles under Minisci Reaction Conditions
1.1 Reagents: Potassium persulfate Solvents: Acetonitrile; 6 h, 80 °C; 80 °C → rt		By: Laha, Joydev K.; et al
1.2 Reagents: Sodium carbonate Solvents: Water; neutralized		Organic Letters (2020), 22(4), 1442-1447.

Scheme 44 (1 Reaction)

Steps: 1 Yield: 90%



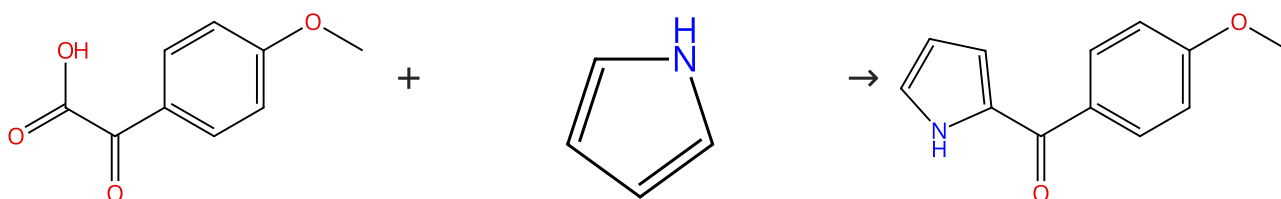
Suppliers (4)

Suppliers (50)

31-614-CAS-39352064	Steps: 1 Yield: 90%	Minisci-Type Carbamoylation of Azauracils with Oxamic Acids
1.1 Reagents: Ammonium persulfate Solvents: Dimethyl sulfoxide; 12 h, 80 °C		By: Pan, Changduo; et al
Experimental Protocols		European Journal of Organic Chemistry (2024), 27(7), e202301177.

Scheme 45 (1 Reaction)

Steps: 1 Yield: 90%



Suppliers (55)

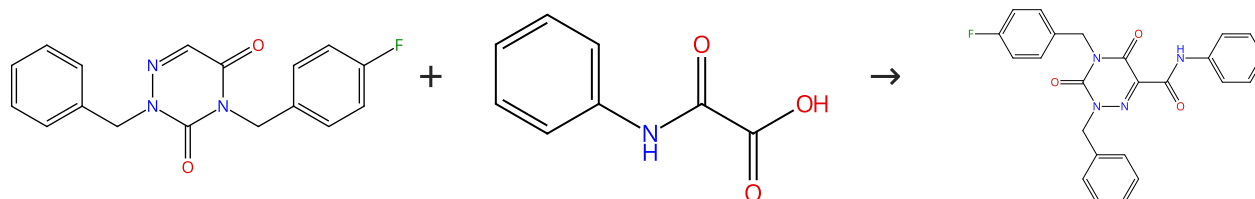
Suppliers (75)

Suppliers (9)

31-219-CAS-21552248	Steps: 1 Yield: 90%	Aroylation of Electron-Rich Pyrroles under Minisci Reaction Conditions
1.1 Reagents: Potassium persulfate Solvents: Acetonitrile; 6 h, 80 °C; 80 °C → rt		By: Laha, Joydev K.; et al
1.2 Reagents: Sodium carbonate Solvents: Water; neutralized		Organic Letters (2020), 22(4), 1442-1447.

Scheme 46 (1 Reaction)

Steps: 1 Yield: 90%

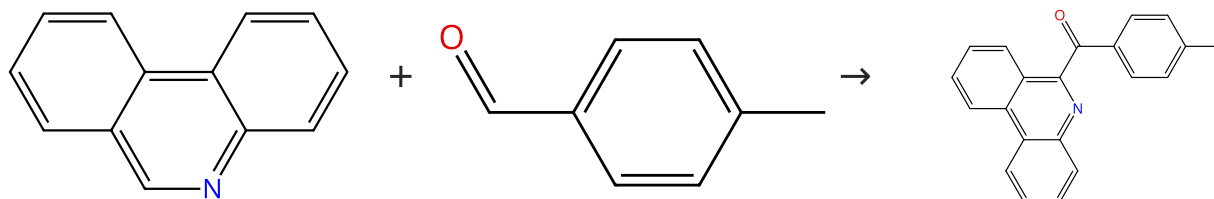


Suppliers (50)

31-614-CAS-39352063	Steps: 1 Yield: 90%	Minisci-Type Carbamoylation of Azauracils with Oxamic Acids
1.1 Reagents: Ammonium persulfate Solvents: Dimethyl sulfoxide; 12 h, 80 °C		By: Pan, Changduo; et al
Experimental Protocols		European Journal of Organic Chemistry (2024), 27(7), e202301177.

Scheme 47 (1 Reaction)

Steps: 1 Yield: 90%



Suppliers (61)

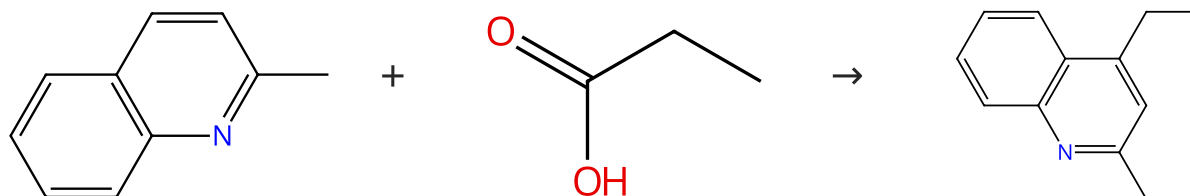
Suppliers (103)

Supplier (1)

31-219-CAS-1676626	Steps: 1 Yield: 90%	Transition metal-free Minisci reaction promoted by NCS and T BHP: acylation of heteroarenes
1.1 Reagents: <i>tert</i> -Butyl hydroperoxide, Chlorosuccinimide Solvents: 1,2-Dichloroethane, Decane; 12 h, 105 °C		By: Siddaraju, Yogesh; et al
Experimental Protocols		Tetrahedron (2016), 72(7), 959-967.

Scheme 48 (1 Reaction)

Steps: 1 Yield: 90%



Suppliers (72)

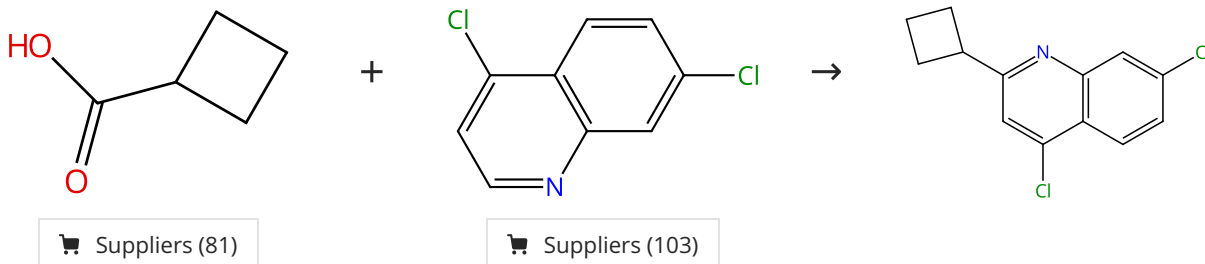
Suppliers (90)

Suppliers (4)

31-614-CAS-36820352	Steps: 1 Yield: 90%	Ag-Catalyzed, Selectfluor-Mediated External Acid-Free Minisci Reaction in Aqueous Solution
1.1 Reagents: Selectfluor Catalysts: Silver nitrate Solvents: Water; 4 h, 80 °C		By: Wang, Wengui; et al
1.2 Reagents: Sodium bicarbonate Solvents: Water		ChemistrySelect (2023), 8(18), e202300958.
Experimental Protocols		

Scheme 49 (1 Reaction)

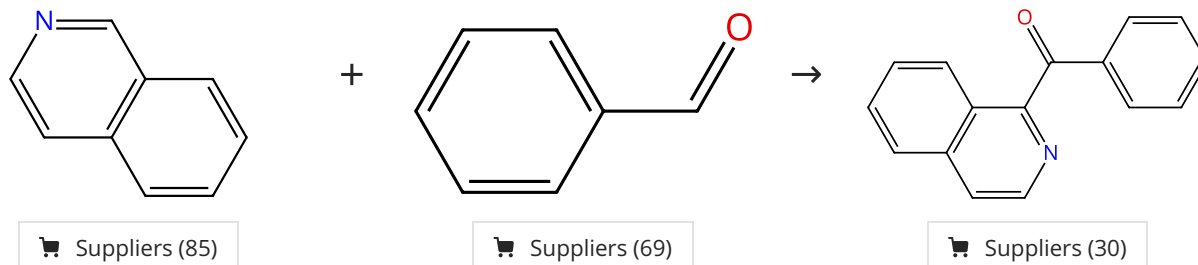
Steps: 1 Yield: 90%



31-219-CAS-19039932	Steps: 1 Yield: 90%	Photoredox-mediated Minisci alkylation of N-heteroarenes using carboxylic acids and hypervalent iodine
1.1 Reagents: 1-(Acetyloxy)-1,2-benziodoxol-3(1 <i>H</i>)-one Catalysts: Tris(2,2'-bipyridyl)ruthenium(II) chloride Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 6 h, 30 °C		By: Wang, Junhua; et al
1.2 Reagents: Potassium carbonate; 5 min		Asian Journal of Organic Chemistry (2018), 7(7), 1307-1310.
Experimental Protocols		

Scheme 50 (1 Reaction)

Steps: 1 Yield: 90%



31-219-CAS-3389410	Steps: 1 Yield: 90%	Regiospecific Benzoylation of Electron-Deficient N-Heterocycles with Methylbenzenes via a Minisci-Type Reaction
1.1 Reagents: <i>tert</i> -Butyl hydroperoxide Catalysts: Aluminum chloride Solvents: Decane; 18 h, 95 °C		By: Ali, Wajid; et al
Experimental Protocols		Journal of Organic Chemistry (2015), 80(11), 5625-5632.